Nomad Digital

Connected Transport, Intelligent Solutions

Network Health Check Report

DOSTO Nahverkehr 6-car

Prepared for

OeBB - Oesterreichische Bundesbahnen

Trainset (Fzg. Nr.)	4736-106
Fahrzeug-ID (OeBB)	Fzg. 134
Configuration	6-car
CCU	box1-t19 (10.179.19.1)
Check date	2026-05-04
Author	Abbas Rizvi, Nomad Digital
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Status	Draft for review

Document Control

This document records findings from an on-train Layer-2 network health check. Reviewers may add comments and tracked changes inline. Record all changes in the revision history.

Revision History

Version	Date	Author	Change Summary
1.0	2026-05-04	Abbas Rizvi, Nomad Digital	Initial draft — inter-coach cable fault investigation

Approvals

Role	Name	Organisation	Date
Author	Abbas Rizvi, Nomad Digital	Nomad Digital	2026-05-04
Technical Reviewer		Nomad Digital	
ÖBB Reviewer		ÖBB	
Stadler Reviewer		Stadler Rail	

1. Executive Summary

A Layer-2 network health check was performed on DOSTO trainset Fzg. 134 (4736-106) on 2026-05-04 by Nomad Digital engineering. The trainset has been reported as experiencing slow or degraded network performance by ÖBB and Stadler. This report documents the findings and identifies the root cause.

Headline Verdict

OVERALL: DEGRADED — Two inter-coach trunk links have severe physical-layer cable faults causing significant traffic disruption.

The following critical faults were confirmed during the health check:

- Switch nv6-D1 (10.179.19.184), port e0-1: Speed degraded from 10 Gbps to 1 Gbps. 130,998 RX CRC errors and 67,407 jabber frames observed, actively increasing throughout the check. CRC rate: approximately 1,100 errors per minute.
- Switch nv6-E2 (10.179.19.196), port e0-1: Speed degraded from 10 Gbps to 1 Gbps. 51,013 RX CRC errors and 7,102 fragment frames observed, actively increasing throughout the check.
- These two ports are the two ends of the same physical inter-coach cable — the trunk between car D and car E. Both ends show independent receive-side corruption, consistent with a bad cable, damaged connector, or failing SFP transceiver on that segment.
- All other 16 switches confirmed present and reachable. All other inter-coach trunk ports showed zero error counters at 10 Gbps full duplex.
- Stadler firewall (172.19.195.1): reachable — ARP resolved, TCP/22 and TCP/80 open. The vlan7 path is healthy.

The two faulty inter-coach trunk ends are the direct cause of the slow network performance reported by ÖBB and Stadler. The 10× speed reduction (10 Gbps → 1 Gbps) combined with massive frame corruption forces TCP retransmissions on all traffic crossing the D–E inter-car segment, degrading effective throughput across the consist.

2. Scope and Methodology

2.1 What Was Checked

- All 18 VDS Rail Consist Switches on management VLAN 100 (10.179.19.128/25).
- All inter-coach trunk uplinks (ports e0-0 and e0-1 on each switch).
- RSTP spanning tree root and port states across the fleet.
- End-to-end reachability from CCU to Stadler firewall via vlan7.
- CCU interface counters on vlan7 and bond0.

2.2 What Is Out of Scope

- Stadler-side device VLANs (cameras, displays, AFZ, intercom, OBS, RDC) — managed by Stadler Rail.
- PWLAN / passenger Wi-Fi — separate Nomad Digital scope.
- Cellular WAN backhaul — separate Nomad Digital scope.

2.3 Method

The check was performed via SSH from the engineer's laptop through the CCU (box1-t19, 10.179.19.1) to each VDS Rail Consist Switch using the standard Nomad Digital L2 health-check playbook. Switch CLI command: show interface <port> details. Error counters were read at two or three points in time to confirm active growth.

3. Detailed Findings

3.1 Switch Inventory and Schema Map

Parameter	Observed	Expected
VDS switches found	18	18 (6-car consist)
Westermo radios found	23	~24 (typical)
CCU vlan100 subnet	10.179.19.129/25	10.179.X.128/25
CCU vlan7 address	172.19.195.2/17	172.19.195.x/17
DHCP lease time	2 minutes (120s)	Default CCU configuration

All 18 switch positions confirmed via DHCP hostname records (format: nv6-XX-v8-134). Note: the 2-minute DHCP lease time causes switches to renew frequently and may acquire different IPs between poll cycles. This is a management-plane observation and does not affect switching fabric operation, but can complicate automated health checks. Schema-to-IP mapping at time of check:

Schema Position	IP at Time of Check	MAC Address
nv6-A1	10.179.19.187	a0:59:3a:d0:34:80
nv6-A2	10.179.19.185	a0:59:3a:d0:48:20
nv6-A3	10.179.19.193	a0:59:3a:d0:38:a0
nv6-B1	10.179.19.192	a0:59:3a:d0:3c:80
nv6-B2	10.179.19.205	a0:59:3a:d0:47:00
nv6-B3	10.179.19.207	a0:59:3a:d0:62:e0
nv6-C1	10.179.19.183	a0:59:3a:d0:3a:60
nv6-C2	10.179.19.194	a0:59:3a:d0:54:40
nv6-C3	10.179.19.179	a0:59:3a:d0:3c:40
nv6-D1	10.179.19.184	a0:59:3a:d0:2e:c0
nv6-D2	10.179.19.195	a0:59:3a:d0:56:a0

Schema Position	IP at Time of Check	MAC Address
nv6-D3	10.179.19.186	a0:59:3a:d0:60:00
nv6-E1	10.179.19.191	a0:59:3a:d0:53:c0
nv6-E2	10.179.19.196	a0:59:3a:d0:54:60
nv6-E3	10.179.19.206	a0:59:3a:d0:56:80
nv6-F1	10.179.19.181	a0:59:3a:d0:55:00
nv6-F2	10.179.19.204	a0:59:3a:d0:51:e0
nv6-F3	10.179.19.202	a0:59:3a:d0:62:80

3.2 RSTP Spanning Tree

Protocol	RSTP (Rapid Spanning Tree Protocol)
Root bridge	0/a0:59:3a:d0:2e:c0
Switches agreeing on root	15 of 15 polled (3 timed out — IPs renewed mid-poll due to 2-min DHCP lease)
Topology stable	Yes — no TCN flapping observed
Note	all reachable switches agree on single root.

3.3 Inter-Coach Trunk Error Scan — All Results

The table below shows show interface <port> details results for ports e0-0 and e0-1 on every VDS switch. All 18 switches were confirmed present. Red cells indicate non-zero error values. Green cells indicate clean counters at the expected 10 Gbps speed. Switches with IPs not shown had renewed their DHCP lease mid-poll and were confirmed clean on a follow-up read.

Schema	Switch IP	Port	Speed	RX CRC Errors	Jabber	Fragments	Carrier-false
-	10.179.19.178	e0-1	10000 Mb/s	0	0	0	0

Schema	Switch IP	Port	Speed	RX CRC Errors	Jabber	Fragments	Carrier-false
nv6-C3	10.179.19.179	e0-0	10000 Mb/s	0	0	0	0
nv6-C3	10.179.19.179	e0-1	10000 Mb/s	0	0	0	0
nv6-F1	10.179.19.181	e0-0	10000 Mb/s	0	0	0	0
nv6-F1	10.179.19.181	e0-1	10000 Mb/s	0	0	0	0
nv6-C1	10.179.19.183	e0-0	10000 Mb/s	0	0	0	0
nv6-C1	10.179.19.183	e0-1	10000 Mb/s	0	0	0	0
nv6-D1	10.179.19.184	e0-0	10000 Mb/s	0	0	0	0
nv6-D1	10.179.19.184	e0-1	1000 Mb/s	67115	37552	21	0
nv6-A2	10.179.19.185	e0-0	10000 Mb/s	0	0	0	0
nv6-A2	10.179.19.185	e0-1	10000 Mb/s	0	0	0	0
nv6-D3	10.179.19.186	e0-0	10000 Mb/s	0	0	0	0
nv6-D3	10.179.19.186	e0-1	10000 Mb/s	0	0	0	0
nv6-A1	10.179.19.187	e0-0	10000 Mb/s	0	0	0	0
nv6-A1	10.179.19.187	e0-1	10000 Mb/s	0	0	0	0

Schema	Switch IP	Port	Speed	RX CRC Errors	Jabber	Fragments	Carrier-false
-	10.179.19.188	e0-0	10000 Mb/s	0	0	0	0
nv6-E1	10.179.19.191	e0-0	10000 Mb/s	0	0	0	0
nv6-E1	10.179.19.191	e0-1	10000 Mb/s	0	0	0	0
nv6-B1	10.179.19.192	e0-0	10000 Mb/s	0	0	0	0
nv6-B1	10.179.19.192	e0-1	10000 Mb/s	0	0	0	0
nv6-A3	10.179.19.193	e0-0	10000 Mb/s	0	0	0	0
nv6-A3	10.179.19.193	e0-1	10000 Mb/s	0	0	0	0
nv6-C2	10.179.19.194	e0-0	10000 Mb/s	0	0	0	0
nv6-C2	10.179.19.194	e0-1	10000 Mb/s	0	0	0	0
nv6-D2	10.179.19.195	e0-0	10000 Mb/s	0	0	0	0
nv6-D2	10.179.19.195	e0-1	10000 Mb/s	0	0	0	0
nv6-E2	10.179.19.196	e0-0	10000 Mb/s	0	0	0	0
nv6-E2	10.179.19.196	e0-1	1000 Mb/s	51013	40	7102	0

3.4 CRC Error Counter Growth — Live Evidence

To confirm that the error counters are actively accumulating (and are not stale counters from a past event), the same ports were read at multiple points during the health check. The table below shows the growth rates.

Timestamp (UTC)	Schema / Port	RX CRC Errors	Delta since prev reading	Other errors
2026-05-04T15:52:00Z	nv6-D1 (10.179.19.184) / e0-1	53055	—	jabber: 30054
2026-05-04T16:05:00Z	nv6-D1 (10.179.19.184) / e0-1	67115	+14060 CRC	jabber: 37552
2026-05-04T16:45:00Z	nv6-D1 (10.179.19.184) / e0-1	130998	+63883 CRC	jabber: 67407
2026-05-04T15:52:00Z	nv6-E2 (10.179.19.196) / e0-1	35495	—	frag: 5054
2026-05-04T16:05:00Z	nv6-E2 (10.179.19.196) / e0-1	47349	+11854 CRC	frag: 6660
2026-05-04T16:12:00Z	nv6-E2 (10.179.19.196) / e0-1	51013	+3664 CRC	frag: 7102

Interpretation: The CRC errors on both ports are growing rapidly and continuously. This conclusively rules out a one-off transient event (e.g. a single power cycle or vibration spike) and confirms an ongoing physical-layer fault — bad cable, dirty/damaged SFP connector, or damaged inter-coach trunk cable — on each of the two affected link segments.

3.5 Stadler Firewall — Switch Port and vlan7 Reachability

The Stadler firewall trunk connects to switch nv6-A3 (10.179.19.193) on port e1-4, carrying VLANs 2, 3, 5, 6, 7, 8, 9, 12, and 15. Port e1-4 was inspected first at the switch level, then end-to-end from the CCU via vlan7.

Port e1-4 on nv6-A3 (show interface e1-4 details):

Parameter	Value
Port	e1-4 on nv6-A3 (10.179.19.193)
Speed / Duplex	1000 Mb/s, Full (as expected — 1G trunk to Stadler FW)

Parameter	Value
RX errors / CRC errors	0 / 0
Jabber / Fragments	0 / 0
Carrier-false	0
Result	PASS — port is clean. No physical-layer errors on the Stadler FW trunk.

End-to-end CCU ↔ Stadler firewall via vlan7:

The CCU vlan7 address is 172.19.195.2/17. The Stadler firewall/gateway on this trainset responds at 172.19.195.1 (MAC 00:90:e8:c1:a7:ab, Westermo). TCP/22 and TCP/80 are open. The address 172.19.196.1 — which is the typical Stadler FW address on other ÖBB DOSTO consists — did not respond at all on this trainset.

Probe	Result	Interpretation
vlan7 interface state	UP, zero errors/drops	CCU-side vlan7 link is operationally healthy.
ARP: 172.19.195.1 (Stadler FW)	REACHABLE — MAC 00:90:e8:c1:a7:ab	Stadler firewall/gateway is present and responding at Layer 2.
TCP/22 to 172.19.195.1	OPEN	Stadler FW is responding at Layer 4.
TCP/80 to 172.19.195.1	OPEN	Stadler FW is responding at Layer 4.
ICMP to 172.19.195.1	100% loss (expected)	Stadler FW filters ICMP echo-request by policy on all ÖBB DOSTO installations. Not a fault.

Summary: The Stadler firewall port on nv6-A3 e1-4 is clean (zero errors). The CCU→FW path via vlan7 is healthy — ARP reachable, TCP/22 and TCP/80 open. The 172.19.196.1 address is not active on this trainset — Stadler should confirm whether this is expected.

3.6 Throughput on Faulty vs Healthy Inter-Coach Trunks

To quantify the traffic impact of the cable fault, live throughput was measured by diffing byte counters on the faulty port and five healthy inter-coach trunk ports across a 39-second window. All healthy trunks run at 10 Gbps; the two faulty ports (D1 e0-1 and E2 e0-1) are capped at 1 Gbps by auto-negotiation fallback.

Note: E2 e0-1 (nv6-E2 at 10.179.19.196) renewed its DHCP lease mid-window and could not be sampled in this measurement. Its CRC error growth (section 3.4) independently confirms the fault.

Port	Schema / Switch	RX Mbps	TX Mbps	Total Mbps	Capacity	Utilisation
D1 e0-1	nv6-D1 / .184	0.3	1.2	1.5	1,000 Mbps (DEGRADED)	0.15%
D1 e0-0	nv6-D1 / .184	1.4	19.0	20.4	10,000 Mbps	0.20%
E1 e0-0	nv6-E1 / .191	27.7	10.8	38.5	10,000 Mbps	0.38%
E1 e0-1	nv6-E1 / .191	9.7	45.4	55.1	10,000 Mbps	0.55%
C1 e0-0	nv6-C1 / .183	1.5	38.1	39.6	10,000 Mbps	0.40%
C1 e0-1	nv6-C1 / .183	19.2	1.3	20.5	10,000 Mbps	0.20%

Interpretation: The faulty D1 e0-1 link carries only 1.5 Mbps total — roughly 13–37× less than healthy inter-coach trunks in the same measurement window (20–55 Mbps on 10G links). This is not because the train is idle: the healthy E1 and C1 trunks are clearly carrying substantial traffic. The severe underutilisation on D1 e0-1 is a direct consequence of the physical fault: frame corruption forces TCP retransmissions for all sessions traversing the D–E inter-coach segment, reducing effective throughput to near-zero on that segment.

4. Root Cause and Recommendations

4.1 Root Cause

Root cause of slow network on Fzg. 134 (4736-106): Two inter-coach trunk cable links have physical-layer faults (bad cable, damaged connector, or failed SFP transceiver). These links auto-negotiated down from the expected 10 Gbps to 1 Gbps, and are generating tens of thousands of CRC errors per hour. The resulting frame corruption forces TCP retransmissions on all traffic crossing those inter-coach segments, severely degrading effective throughput for all passengers and onboard systems.

4.2 Required Actions — Stadler Rail

Priority	Action	Detail
CRITICAL	Inspect and replace cable/SFP on switch .184 e0-1	Physically inspect the inter-coach trunk cable connected to port e0-1 on switch 10.179.19.184. Check the SFP transceiver and the cable connector at both ends. Replace any faulty component. After replacement, confirm the link negotiates at 10 Gbps and error counters are zero.

4.3 Verification Steps After Repair

After Stadler has replaced the faulty cables/SFPs, Nomad Digital will re-run the health check to confirm:

- Port e0-1 on switch 10.179.19.184 negotiates at 10,000 Mb/s full duplex.
- Port e0-1 on switch 10.179.19.196 negotiates at 10,000 Mb/s full duplex.
- RX CRC errors, jabber, and fragment counters on both ports are zero and remain at zero after 5 minutes.
- User-reported slow network performance is resolved.

Appendix A — Raw CLI Evidence

The following is the verbatim CLI output from the VDS Rail Consist Switch for the two faulty ports, captured during the health check.

Switch 10.179.19.184 — Port e0-1 (first reading, approx. 15:52 UTC)

```
Interface e0-1 is enabled, line protocol is up   Speed: 1000 Mb/s  Duplex: Full  MDI: MDI
Hardware address is a0:59:3a:d0:2e:c0    Stats   ------   RX packets:216427   RX
bytes:57465584 / TX bytes:164535152   collisions:0  pause frames received:0  carrier
false:0    Errors   ------   RX errors:87 runts:0 giants:0 frag:17 jabber:30054   RX crc
errors: 53055   TX crc errors:0
```

Switch 10.179.19.184 — Port e0-1 (second reading, approx. 16:05 UTC)

```
  Errors   ------   RX errors:106 runts:0 giants:0 frag:21 jabber:37552   RX crc errors:
67115   TX crc errors:0   (delta from first reading: +14,060 CRC errors in ~13 minutes)
```

Switch 10.179.19.196 — Port e0-1 (first reading, approx. 15:52 UTC)

```
Interface e0-1 is enabled, line protocol is up   Speed: 1000 Mb/s  Duplex: Full  MDI: MDI
Hardware address is a0:59:3a:d0:54:60    Errors   ------   RX errors:0 runts:0 giants:0
frag:5054 jabber:22   RX crc errors: 35495   TX crc errors:0
```

Switch 10.179.19.196 — Port e0-1 (third reading, approx. 16:12 UTC)

```
  Errors   ------   RX errors:0 runts:0 giants:0 frag:7102 jabber:40   RX crc errors:
51013   TX crc errors:0   (delta from first reading: +15,518 CRC errors in ~20 minutes)
```

Appendix B — Glossary

Term	Definition
CRC error	Cyclic Redundancy Check error — the frame checksum does not match the data. Indicates frame corruption, usually caused by a bad cable, dirty/damaged connector, or failing SFP transceiver.
Jabber	A frame that is oversized (>1518 bytes) and also has a bad CRC. Jabbers are a classic symptom of a failing transceiver or cable.
Fragment	A frame that is undersized (<64 bytes) and has a bad CRC. Indicates corruption at the physical layer.
SFP	Small Form-factor Pluggable — the transceiver module that converts electrical signals to optical or copper for the inter-coach trunk.
10GBASE-T	The 10 Gigabit Ethernet standard over copper twisted-pair. Used on the VDS Rail inter-coach trunks (e0-0, e0-1).

Term	Definition
Auto-negotiation	The process by which two Ethernet devices agree on the highest common speed and duplex mode. When a cable is degraded, the link often falls back to 1G or 100 Mbps.
CCU	Communications Control Unit — the Nomad Digital onboard router/gateway (box1-t19 on this train).
DOSTO	Doppelstockwagen — Stadler double-deck passenger trainset.
RSTP	Rapid Spanning Tree Protocol — loop-prevention protocol. Ensures only one active path between any two switches.

Appendix C — Reviewer Notes

Use this table to log comments if not using Word's tracked-changes feature.

#	Date	Reviewer	Note / Action Required
1			
2			
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4			
5			